

Charge transport in organic materials with quasi-Gaussian density of states

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ABSTRACT

Opto-electronic devices based on organic semiconductors have garnered significant research interest in recent years, with transport energy being a critical factor governing their performance. The density of states (DOS) plays a key role in modulating transport energy within these materials, thereby influencing charge carrier mobility and the Einstein relation. In this work, the carrier transport within a set of quasi-Gaussian DOSs, $g(E) \propto \exp[-(E/\sigma)^p]$ with $1 \leq p \leq 2$, is examined. We systematically investigate the effects of DOS distribution width, power p , carrier concentration, and material parameter $N\alpha^3$ on transport energy. It is observed that at high carrier concentrations, the transport energy exhibits a pronounced dependence on concentration variations. Furthermore, the generalized Einstein relation (GER) and the behavior of mobility are studied by adjusting the lower integration limit of the carrier concentration in GER expression to E_t . Our findings only deviate slightly from the classical Einstein relation. Subsequently, the experimental data of a small-molecule material and a polymer material are fitted, and the power p is determined by using the approach established by Oelerich *et al.* The resulting trend in mobility demonstrates approximately agreement with experimental data. These results offer additional alternatives for this field and provide valuable insights for reference.

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I. INTRODUCTION

Organic semiconductor materials have triggered considerable attention from the academic community owing to their mechanical flexibility, lightweight characteristics, superior optical properties, and compatibility with large-scale fabrication processes. Opto-electronic devices based on these materials exhibit notable commercial viability, with their applications having been extensively deployed in the market.¹ Targeting performance enhancement in organic opto-electronic devices, attention is directed to charge transport, which serves as a pivotal determinant.² While charge mobility and diffusion coefficient are two essential parameters describing the charge transport properties of the materials.^{3,4}

Transport energy (E_t) is a particular energy level which plays a crucial role in hopping transport of carriers in the band tail of various disordered organic and inorganic semiconductors.⁵ The mobility of a material can be derived via transport energy theory combined with the Einstein relation, while charge mobility μ and diffusion coefficient D are related by $D/\mu = kT/q$, with $k = 1.38 \times 10^{-23}$ J/K being the Boltzmann constant, T the temperature, and q the element charge. Meanwhile, the Einstein relation

crucially relies on transport energy to differentiate free charges from trapped charges.⁶ Concurrently, the Einstein relation itself remains a subject of ongoing research interest.^{7,8} Recent research findings indicate that the density of states (DOS) may influence the E_t behavior of materials. Thereby, DOS impacts both charge carrier mobility and the form of the Einstein relationship of these materials.⁹ One aspect is that, in systems with an exponential DOS, charge mobility consistently exhibits a concentration-dependent behavior;¹⁰ In contrast, for systems with the Gaussian DOS, mobility depends on concentration only under high-concentration conditions.¹¹ Furthermore, the Einstein relations corresponding to these two DOS models also differ significantly. At low concentrations, the generalized Einstein coefficient for the exponential DOS is proportional to the width of the DOS, whereas for the Gaussian DOS, this coefficient approximates unity.^{9,12}

Conventional studies of charge transport typically restricted to models based on exponential and Gaussian DOSs. The quasi-Gaussian DOS introduces an additional degree of flexibility, enabling a more accurate description of systems where the actual DOS lies between these two extremes. Given the complexity of the

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actual DOS shapes of organic materials,¹³ research on the quasi-Gaussian DOS can provide significant assistance and facilitate more reliable analyses. Recently, Oelerich proposed an approach for determining the material DOS, which entails calculating the transport energy of the material and subsequently deriving the power p based on the mobility–concentration relationship of the material.^{3,14} However, several key aspects related to the quasi-Gaussian DOS remain unexplored. First, according to the expression of the DOS, since that p is a decimal number, the energy value have to be positive, and the asymmetry distribution seems unrealistic. Second, a systematic research of the variation of transport energy with parameters including the power p , carrier concentration, DOS width, and localization length has not been carried out. Third, the dependence of mobility on carrier concentration has not been analyzed. Finally, the evolution of the Einstein relation with respect to the power p , carrier concentration, and DOS width remained unstudied.

In this work, a symmetric DOS is first constructed by introducing an absolute energy value. Subsequently, the variation patterns of transport energy under different parameter configurations are calculated. Based on these results, the mobility and Einstein relation behaviors are further analyzed, and the obtained theoretical patterns show roughly consistency with experimental observations.

II. THEORY AND RESULTS

A. DOS and Miller–Abrahams rate

In disordered organic systems, the key factor determining the opto-electronic properties of a material is its energy spectrum, also known as the DOS.^{10,15} In organic materials, exponential and Gaussian DOSs are the most commonly adopted types, yet the real DOS may take complex shapes which can be expressed as the superposition of multiple elementary functions. Oelerich suggested a recipe on how to determine the DOS in disordered organic semiconductors from the measured dependence of the charge carrier mobility on the carrier concentration.³ Based on their theory, in this part, we will carry out a systematic investigation to explore the relationship between transport energy and various parameters with different power p values. It is worth noting that the power p limits the energy value to be positive. Therefore, we take the absolute value of the energy.¹⁶ The energies of localized states are distributed via a DOS of the form,⁵

$$g(E) = \frac{N}{C} \exp \left[- \left(\frac{|E|}{\sigma} \right)^p \right], \quad 1 \leq p \leq 2, \quad (1)$$

where N is the total concentration of localized states, p is between 1 and 2 for most organic materials, and σ here differs from Gaussian standard deviation by a factor of $\sqrt{2}$. When $p=1$, that is in the case of exponential DOS, it is assumed that $\sigma = kT_0$ and the typical value of T_0 is usually smaller than 600 K. In the case of Gaussian DOS, i.e., $p=2$, the typical value of σ could be $4\sqrt{2}-5\sqrt{2} \text{ kT}$.^{10,17} Here, we choose negative energy as the higher energy direction. The normalization condition,

$$\int_{-\infty}^{\infty} g(E) dE = 2N, \quad (2)$$

demands $C = \sigma \Gamma(p^{-1} + 1)$.

The hopping process can be described by either the Miller–Abrahams expression or Marcus formula.^{18,19} Since the Miller–Abrahams expression is more extensively utilized in academic and practical contexts, we will employ it in the analysis of E_t behavior. According to the Miller–Abrahams expression, the hopping rate from an electron donor at energy E_i to an electron acceptor with energy E_j at a distance R_{ij} can be described as

$$v_{ij} = v_0 \exp[-2\alpha R_{ij}] \exp \left[- \frac{E_j - E_i}{kT} \theta(E_j - E_i) \right], \quad (3)$$

where v_0 is the attempt-to-escape frequency, α^{-1} is the localization radius of a charge carrier, T is the temperature, and θ is the Heaviside function, which means when $E_j \leq E_i$, the value of this function is 0. Otherwise, the function value takes 1.

B. Transport energy

It is well known that in organic semiconductors with a DOS, as in amorphous inorganic semiconductors, all transport phenomena at low temperatures are determined by hopping of charge carries in the vicinity of a particular energy level, called the transport energy. E_t determines each hopping event of a carrier in the DOS and simplifies the descriptions of all phenomena dominated by the hopping. The exponential and Gaussian cases are extensively investigated,^{15,20} while a systematical research of charge transport in the case of the quasi-Gaussian DOS is yet carried out with the influencing parameters. For a particular DOS, the transport energy can be derived as follows. Given an electron with energy E_i , the median rate of an upward hop to a neighboring localized state with energy $E_t \leq E_i$ is written as

$$v_{\uparrow} = v_0 \exp \left[-2\alpha R(E_t) - \frac{E_i - E_t}{kT} \right], \quad (4)$$

$$R(E_t) = \left[\frac{4\pi}{3} \int_{E_t}^{\infty} dE g(E) \right]^{-\frac{1}{3}}. \quad (5)$$

It should be noted that the typical distance to the target site described in Eq. (5) does not consider the occupation effect of the target site. To extend the consideration to such cases, an analytical model describing the effect of partially filled localized states on the value of E_t based on variable range hopping theory was advanced.¹⁵ Though the work is dedicated to the case of Gaussian DOS, the basic idea is applicable to any forms of DOS. Based on the theory described above, the typical distance to the target site in a hopping event via the concentration of states below the target energy can be calculated as

$$R(E) = \left[\frac{4\pi}{3} \int_{E_t}^{\infty} g(E) [1 - f(E, E_F)] dE \right]^{-\frac{1}{3}}, \quad (6)$$

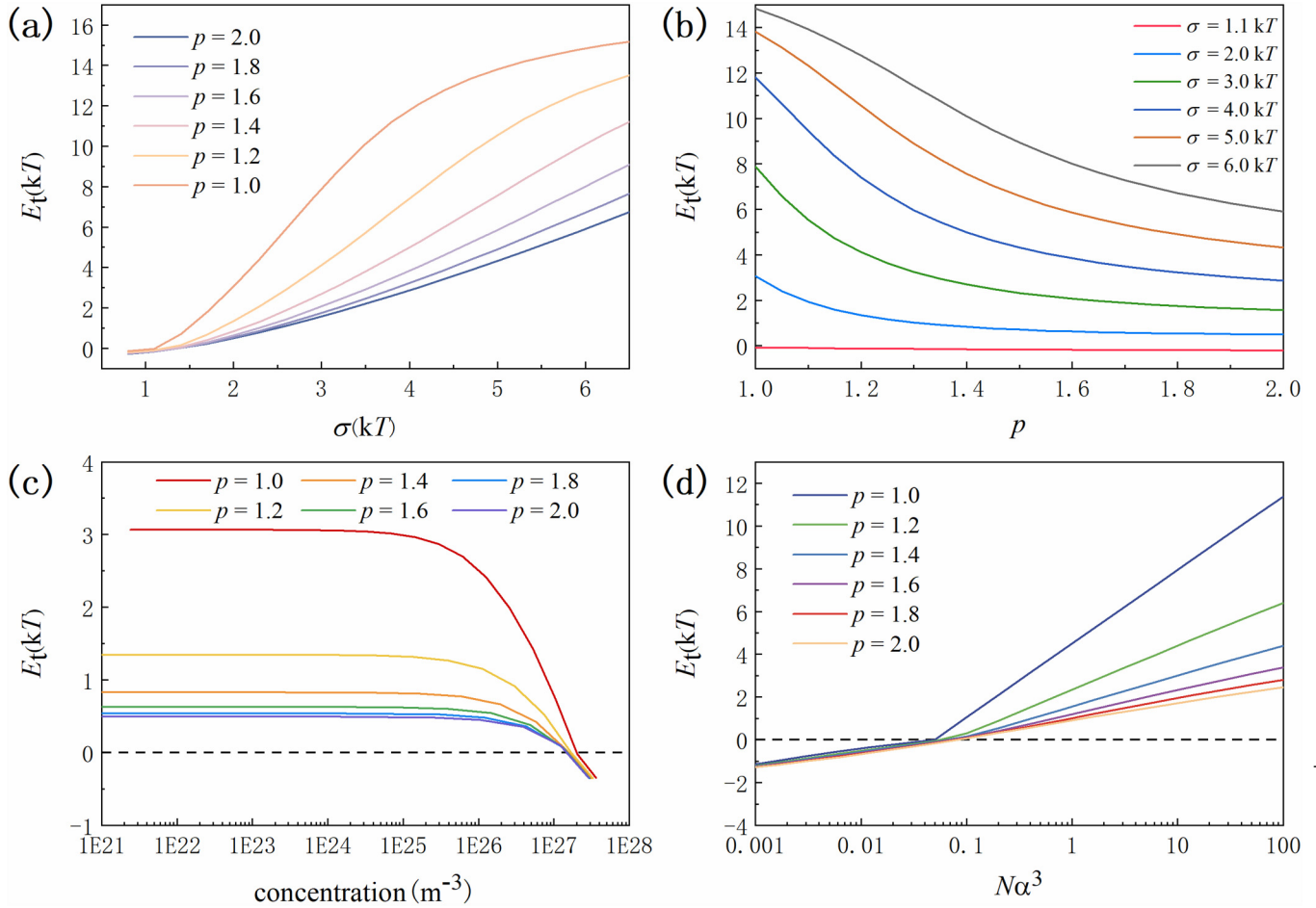


FIG. 1. (a) E_t vs σ curves for $p = 1.0, p = 1.2, p = 1.4, p = 1.6, p = 1.8$, and $p = 2.0$ with $N\alpha^3 = 0.1$ and $E_F = 20$ kT . (b) E_t vs p curves for $\sigma = 1.1$ $kT, \sigma = 2.0$ $kT, \sigma = 3.0$ $kT, \sigma = 4.0$ $kT, \sigma = 5.0$ kT , and $\sigma = 6.0$ kT in the case of $N\alpha^3 = 0.1$ and $E_F = 20$ kT . (c) E_t vs concentration curves for $p = 1.0, p = 1.2, p = 1.4, p = 1.6, p = 1.8$, and $p = 2.0$ with $\sigma = 2$ kT and $N\alpha^3 = 0.1$. (d) E_t vs $N\alpha^3$ curves for $p = 1.0, p = 1.2, p = 1.4, p = 1.6, p = 1.8$, and $p = 2.0$ with $\sigma = 2$ kT and $E_F = 20$ kT .

where $f(E, E_F) = 1/\{1 + \exp[(E - E_F)/kT]\}$ is the Fermi function describes the energy distribution of carriers and E_F is the Fermi energy. By inserting Eq. (6) into Eq. (4) and subsequently maximizing the upward hopping rate $v_{\uparrow}$ via

$$\frac{\partial v_{\uparrow}(E_i, E_t)}{\partial E_t} = 0, \quad (7)$$

E_t can be solved from the following transcendental equation:

$$\frac{2}{3} \left(\frac{4\pi}{3B} \right)^{-\frac{1}{3}} \frac{kT}{a} \left[\int_{E_i}^{\infty} [1 - f(E, E_F)] g(E) dE \right]^{-\frac{4}{3}} \times [1 - f(E_t, E_F)] g(E_t) = 1, \quad (8)$$

where B is the average number of available sites for any hopping event and $B = 2.7$.

Simulations of the influences of relevant parameters on E_t are displayed in Fig. 1. Generally, in the case of Gaussian distribution, the typical value of σ is in the range of $4\sqrt{2}$ – $6\sqrt{2}$ kT , while in the case of exponential distribution, the value of σ is usually less than 2 kT .^{2,17} For exponential defect state distributions, cases where σ exceeds 2 kT , and even approaches 4 kT , have been reported.²¹ For narrow distribution with $\sigma \leq 1$ kT and broad distribution with $\sigma > 6$ kT , they are rarely found in organic materials, so we will not discuss them any further. As shown in Fig. 1(a), with σ increases from 1 to about 6 kT , the calculated E_t values increase in all these cases, with an initial fast rise followed by a slower increase. For a fixed σ value, DOS with large p -value possesses a relative small E_t . Figure 1(b) gives E_t values as a function of p with $\sigma = 1.1$ $kT, \sigma = 2.0$ $kT, \sigma = 3.0$ $kT, \sigma = 4.0$ $kT, \sigma = 5.0$ kT , and $\sigma = 6.0$ kT . As p -value increases, E_t exhibits a monotonic decrease. Conventionally, researches from the 1990s are commonly limited to low carrier concentrations and a constant value of E_t is usually

supposed for simplicity, which is frequently modeled as a mobility edge. In fact, as Fig. 1(c) shows, E_t is indeed not affected by the carrier concentration at low concentrations. While at high concentrations, E_t has a strong dependence on the concentration. This perspective aligns with the results of our earlier work.⁵ Under high carrier concentration, with a further increase in carrier concentration, the available target energy level lift and across the center of DOS, corresponding to a negative value of E_t . The negative E_t observed under high carrier concentrations reflects the significant upward shift of available hopping sites relative to the reference energy. Indeed, according to Eq. (7), this is a natural outcome, as E_t is a function of E_F . The product $N\alpha^3$ is a dimensionless parameter that represents the average number of localized states contained within a localization volume.²² The dimensionless parameter $N\alpha^3$ serves as intrinsic parameter of charge transport. Even when two materials have different values of N and α , an identical $N\alpha^3$ value suggests that their conduction mechanisms are probably similar. As displayed in Fig. 1(d), when $N\alpha^3$ is less than approximately 0.02, the curves of E_t vs $N\alpha^3$ under different p values almost coincide, and then after the turning point of $N\alpha^3 = 0.02$, the value of E_t begins to increase strongly, and the smaller the value of p , the greater the increase in E_t . The emergence of the turning point is due to the energy peak from the absolute value of the energy according to Eq. (1).

C. Generalized Einstein relation

The generalized Einstein relation, that is the ratio between the diffusion coefficient D and the mobility μ , is given as

$$\frac{D}{\mu} = \frac{n}{q \frac{\partial n}{\partial E_F}}, \quad (9)$$

where n is the particle concentration and E_F is as previously defined.^{9,23–25} Another commonly used version of Eq. (9) is that E_F is replaced by the chemical potential in the right side. The distinction between the chemical potential and the Fermi energy E_F will not be discussed here, as they are approximately equal at the temperatures considered,

$$\frac{D}{\mu} = \frac{kT}{q} \frac{\int \exp\left(-\left(\frac{E}{\sigma}\right)^p\right) \frac{1}{1 + \exp(E - E_F)} dE}{\int \exp\left(-\left(\frac{E}{\sigma}\right)^p\right) \frac{\exp(E - E_F)}{(1 + \exp(E - E_F))^2} dE} = \frac{kT}{q\eta}. \quad (10)$$

Here, we denote the second fraction by the coefficient η for simplicity.

In 2002, to compute the effect of a disordered density of states on the diffusion-mobility ratio, Roichman *et al.* chose the upper and lower bounds of the integral as positive and negative infinity and explained the deviation from the classical Einstein relation.⁹ However, in 2006, by assuming the low-energy carriers do not contribute to carrier mobility, Neumann *et al.* examined the scenario of trap-controlled transport and adjusted the lower bound of the integral to the mobility edge, it was found that the extent of this deviation mitigated.²⁶ It is interesting that, to discuss the validity of Einstein relation in organic semiconductors, Li *et al.* selected the

integration interval ranging from E_F to E_t . According to the percolation model, only states within a fixed energy range around E_F contribute significantly. Thus, selecting E_F as the lower integration limit of carrier concentration is justified under specific conditions. Nevertheless, the charges located above the E_t level also contribute to the conduction process, making the choice of integration limit somewhat contentious. In this work, to investigate the Einstein relation under variations of the power p , the two scenarios that using E_F and E_t as the lower integration limit of carrier concentration are considered, respectively. The choice of negative infinity as the upper integration limit of carrier concentration is based on the assumption that no particular restrictions are imposed on states lying above E_t , consistent with the approach adopted in the 2006 study.²⁶ While in 2006, the value of E_t was not explicitly given and was tentatively set to zero for simplification. In contrast, our approach allows for the determination of E_t value. To achieve more accurate results, we select E_t as the lower integration limit of carrier concentration as one case. Given that E_F indicates the charge filling level, it can also be reasonably chosen as the lower integration limit of carrier concentration.

As shown in Fig. 2(a), if we set the integral range as from E_F to negative infinity in Eq. (10), the value of μ/D changes relative slowly at low concentrations, but it significantly deviates from the classic Einstein relation as the concentration increases. The findings of our work indicate that when $p = 1$, the value of μ/D converges to 0.25. This result aligns precisely with theoretical analysis considering the drift carrier mobility and the diffusion coefficient separately reported by Baranovskii *et al.*¹² At the same time, the findings are later proved by the experimental data presented in Refs. 27 and 28. As displayed in Fig. 2(b), we set the integral range from E_t to negative infinity in Eq. (10) and the value of E_t could be calculated, the extent of deviation of classic Einstein relation decreases. Moreover, we find that there is an abrupt change in the μ/D value at high concentrations in Fig. 2(b), which was caused by a decrease in the integral interval of carrier concentration. This concentration range exceeds the carrier concentration in commonly used devices, and its correction will not be discussed in our work. It is observed that a small value of p leads to a more significant deviation in the curves, a trend that is more pronounced in the case where E_F serves as the lower integration limit. A more rigorous treatment for choosing the lower integration limit, which involves the consideration of charge transit time, will be addressed in a separate work.

D. Determination of p and mobility fitting

Following the theoretical procedure of Baranovskii *et al.*, the concentration-dependent mobility in disordered semiconductors is written as³

$$\mu_0 = \frac{q}{kT} R^2(E_t) t^{-1}, \quad (11)$$

where the averaging hopping time is the weight over all the upward hop events,

$$t = \frac{\int_{E_t}^{\infty} v_{\uparrow}^{-1}(E_t) [1 - f(E)] g(E) dE}{\int_{E_t}^{\infty} [1 - f(E)] g(E) dE}. \quad (12)$$

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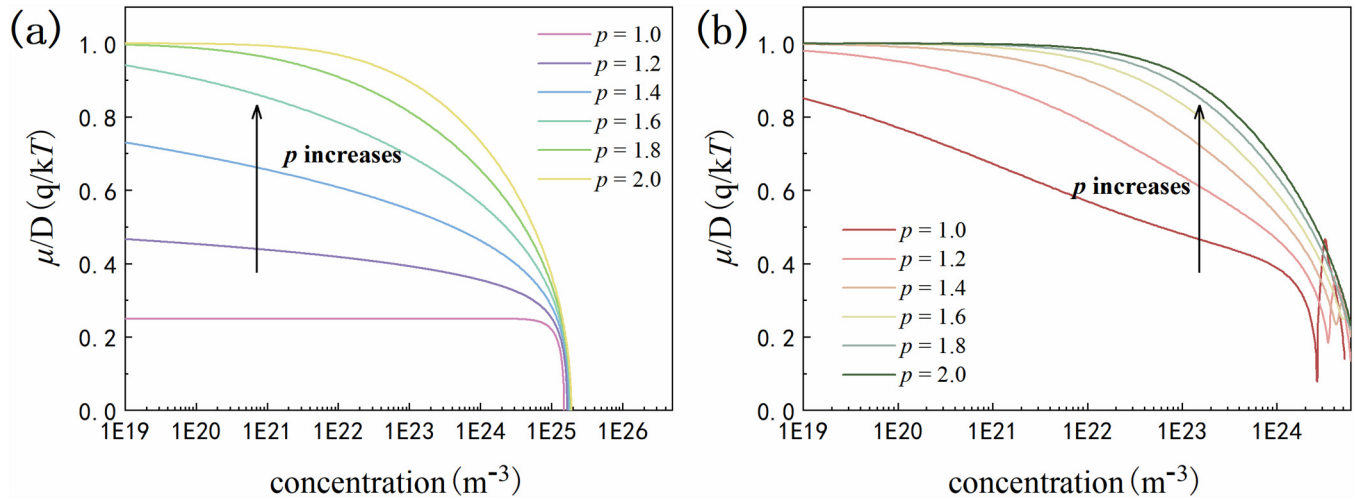


FIG. 2. The inverse of Einstein relation (μ/D) vs the carrier concentration for different values of p with $\sigma = 2.828$ kT. (a) The integral range is from E_F to negative infinity in Eq. (10). (b) The integral range is from E_t to negative infinity in Eq. (10).

From Eqs. (7), (10), and (11), we can straightforwardly obtain the equation for the carrier mobility,

$$\mu_0 = v_0 \frac{q}{kT} \frac{3B}{4\pi R(E_t)n_t} \exp\left(\frac{-2R(E_t)}{\alpha} - \frac{E_F - E_t}{kT}\right), \quad (13)$$

with n_t determined as

$$n_t = \int_{E_t}^{\infty} f(E, E_F)g(E)dE. \quad (14)$$

When we consider the generalized Einstein relation, the mobility equals Eq. (11) multiplied by the coefficient η ,

$$\mu = \eta \frac{q}{kT} R^2(E_t)t^{-1}. \quad (15)$$

Thus, through calculation, the generalized carrier mobility can be expressed as

$$\mu = v_0 \eta \frac{q}{kT} \frac{3B}{4\pi R(E_t)n_t} \exp\left(\frac{-2R(E_t)}{\alpha} - \frac{E_F - E_t}{kT}\right). \quad (16)$$

To illustrate the use of our model, we simulated the relationship between μ and carrier concentration for a representative small-molecule material and a polymeric material.^{29,30} In order to determine the value of p used, we adopt the recipe from Oelerich *et al.*³ and it was determined that μ is independent of the concentration at low concentrations, the scenario in which $p < 1.5$ could be excluded, a detailed explanation will be given in the following part.

Then, when the carrier concentration increases sufficiently for the Fermi level E_F to rise and align with the equilibrium energy

E_{∞} ,

$$E_{\infty} = \frac{\int_0^{\infty} E g(E) \exp\left(\frac{E}{kT}\right) dE}{\int_0^{\infty} g(E) \exp\left(\frac{E}{kT}\right) dE}. \quad (17)$$

μ begins to exhibit a pronounced dependence on carrier concentration. For the case of $p < 1.5$, the equilibrium energy E_{∞} is situated at very deep energy levels. As a result, the relaxation time for carriers to achieve thermal equilibrium becomes extremely long, far exceeding typical experimental timescales. This is inconsistent with the observation of steady-state, non-dispersive transport. Consequently, the regime $p < 1.5$ is excluded from consideration. The carrier density at which this transition occurs is defined as the critical concentration n_c , which is, thus, determined by the condition,

$$E_F(n_c) = E_{\infty}, \quad (18)$$

where the Fermi level E_F determined by the condition,

$$\int_{-\infty}^{\infty} f(E, E_F)g(E)dE = n_c. \quad (19)$$

For copper phthalocyanine (CuPc) and *N*-alkyl ring-substituted poly(p-phenylene vinylene) (NRS-PPV), the width of DOS is about 4 kT, corresponding to 0.1 $\sqrt{2}$ eV in Eq. (1), and a range of values around it is explored. Upon acquiring the n_c value, a relationship between relative concentration n/N and p in the cases of $\sigma = 0.08\sqrt{2}$ eV, $\sigma = 0.1\sqrt{2}$ eV, and $\sigma = 0.1\sqrt{2}$ eV is plotted in Fig. 3(a). Here, n is the charge carrier concentration, and N is the total concentration of localized states. The ratio n/N represents the

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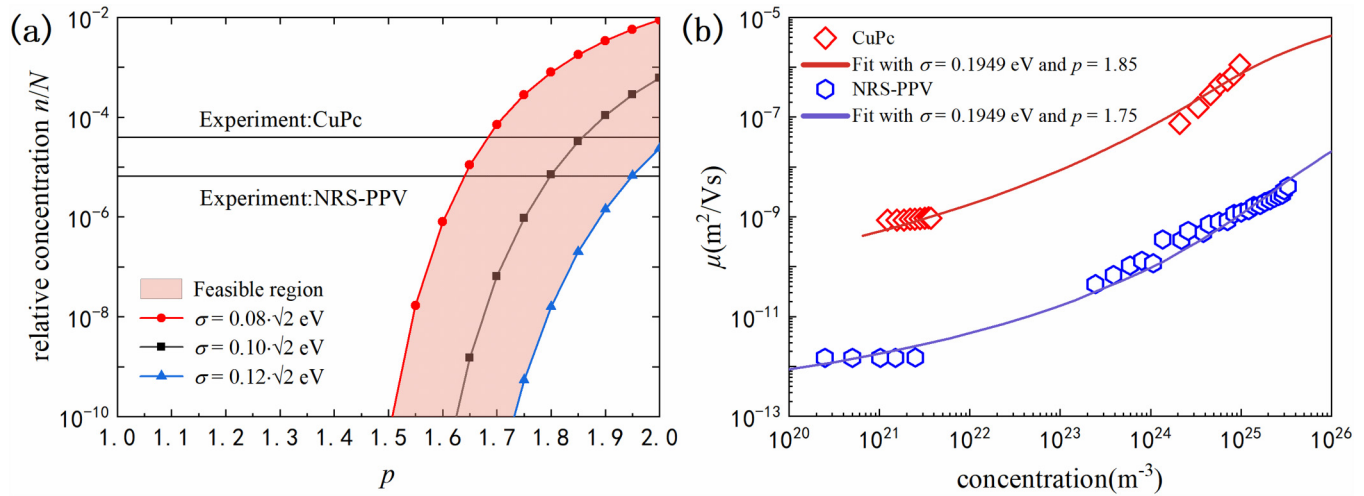


FIG. 3. (a) Relative concentration vs p curves for $\sigma = 0.08\sqrt{2}$ eV, $\sigma = 0.1\sqrt{2}$ eV, and $\sigma = 0.12\sqrt{2}$ eV. The experimental data of CuPc and NRS-PPV are marked. The colored-filled areas in the figure are the feasible region. (b) Fits of experimental data of CuPc and NRS-PPV. The value of p was calculated by adopting the approach described in Ref. 3. μ vs concentration curves for $\sigma = 0.1949$ eV and $p = 1.75$ when fitting the experimental data of CuPc. μ vs concentration curves for $\sigma = 0.1949$ eV and $p = 1.85$ when fitting the experimental data of NRS-PPV. The experimental data for CuPc are reproduced with permission from Sharma *et al.*, J. Phys.: Condens. Matter **28**, 235802 (2016). Copyright 2016 IOP Publishing. The data for NRS-PPV are reproduced with permission from Tanase *et al.*, Phys. Status Solidi A **201**, 1236–1245 (2004). Copyright 2004 Wiley-VCH GmbH.

relative concentration. The filled area corresponds to the feasible region. By utilizing the experimental critical concentration and different values of σ , the range of p is determined from Fig. 3(a). For typical organic materials, the DOS width σ generally lies within 4–5 kT. At room temperature (293 K), kT_0 is approximately 0.025 eV. In Fig. 3(a), the border line with $\sigma = 0.08\sqrt{2}$ eV (red line with solid circles) corresponds to a Gaussian width of about 3 kT, and the region on its left corresponds to narrower distribution widths, which are unattainable for amorphous materials. This observation aligns with the results obtained from the preceding analysis of E_∞ and further supports our exclusion of $p < 1.5$. Subsequently, in order to obtain a more precise p -value, the experimental data are fitted as plotted in Fig. 3(b), depicting the mobility μ and concentrations of CuPc and NRS-PPV.^{2,29,30} Figure 3(b) shows an approximately accordance between our model and the experimental data. For the curve of CuPc, $\sigma = 0.1949$ eV, $p = 1.75$, $N_0 = 8 \times 10^{26} \text{ m}^{-3}$, $\alpha = 6 \times 10^{-10} \text{ m}$, and $v_0 = 2.5 \times 10^{13} \text{ s}^{-1}$ are selected. For the curve of NRS-PPV, $\sigma = 0.1949$ eV, $p = 1.85$, $N_0 = 5 \times 10^{26} \text{ m}^{-3}$, $\alpha = 2 \times 10^{-10} \text{ m}$, and $v_0 = 4.91 \times 10^{14} \text{ s}^{-1}$ are selected. Here, p is regarded as a fitting parameter and the σ values used are consistent with previous simulations.^{25,30} The values of N_0 s are with the same orders of molecule density or unit density of polymer calculated from the material weight densities, i.e., $1.63 \times 10^{27} \text{ m}^{-3}$ for CuPc and $1.21 \times 10^{27} \text{ m}^{-3}$ for NRS-PPV. Systematic studies suggests that v_0 generally lies within 10^{12} – 10^{15} s^{-1} .^{31,32} The molecular dimension a , calculated from the mass density, is approximately $8.5 \times 10^{-10} \text{ m}$ for CuPc and $9.4 \times 10^{-10} \text{ m}$ for NRS-PPV, and α typically chosen smaller than a . Given the specifics of the experimental conditions, Eq. (13) is employed for data fitting, with the aforementioned parameter values being assigned based on practical considerations.^{29,30}

This model is applicable to describing charge transport processes based on hopping mechanism in disordered organic semiconductors characterized by a symmetric quasi-Gaussian density of state distribution. It can be employed to analyze the dependence of transport energy, mobility, and the generalized Einstein relation on material parameters and carrier concentration. Furthermore, the model enables the extraction of microscopic material parameters by fitting experimental data. In order to be able to make a full comparison with the experimental J - V curve for disordered systems, the model should be extended in two directions. First, in the presence of an electric field, the Miller–Abrahams rate requires corresponding modifications, and the charge carrier mobility is consequently affected. Second, for cases involving polarons, the Miller–Abrahams hopping rate should be replaced by the Marcus formula.

III. CONCLUSION

In summary, we have systematically examined how transport energy varies with key parameters. As the power p increases, the influence of the distribution width on transport energy gradually diminishes. At low carrier concentrations, transport energy approaches a constant value, whereas at high concentrations, it exhibits a clear dependence on carrier density. Regarding material parameter $N\alpha^3$, the power p before the turning point has almost no effect on the transport energy, while after the turning point, transport energy decreases as the power p increases. In the generalized Einstein relation, we set E_t as the lower limit for the integral of the coefficient, and the numerical value of E_t can be accurately calculated. This approach substantially reduces deviations from the

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classical Einstein relation. The expression of mobility is derived, and the power p is graphically determined and analyzed using the method established by Oelerich *et al.* The resulting theoretical framework shows approximately consistency with experimental observations. Through the fitting of experimental data, an accurate p -value is obtained.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Siyan Li: Conceptualization (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Validation (equal); Writing – original draft (equal). **Guisong Wang:** Data curation (equal); Validation (equal). **Jiayang Yu:** Investigation (equal); Visualization (equal). **Tianyou Zhang:** Conceptualization (equal); Methodology (equal); Resources (equal); Software (equal); Supervision (equal); Writing – original draft (equal); Writing – review & editing (equal). **Lei Wang:** Project administration (equal); Resources (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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